



Focal Facts

Michael Browne

Tracing breakthroughs in technology has been both, one of our favourite pastimes as well as our solemn duty as harbingers of technology related information. The era of digitalisation is both the present and the future. And this has touched our every day lives too, as fluctuating waves (representatives of the analog era) have given way to zeroes and ones. Vinyl gave way to CDs and later MP3s, video cassettes to VCDs and DVDs, similarly did film eventually take the digital plunge, and it's a path from whence there is no return. And why should there be? The digital photography revolution brought about drastic reductions in the cost of adoption of photography as a hobby while reducing the learning curve and the cost of making mistakes while learning. Sure in the beginning there were teething issues and sceptics wrote off digital photography as an amateur hobby, professionals stuck with film. It's been years since then, and today film is all but dead, restricted to museums and a very few niche professionals who still don't believe.

In what follows, we'll look into one of the most remarkable breakthrough products in modern times—the digital camera. A product that has revolutionised a profession and changed the way people create and share

memories, and simultaneously gained adoption into nearly each and every one of our homes.

It's All About Pixels, Really

The first thing about digital photography is its complete compatibility with anything digital, be it a computer, a cell phone, another digital camera or even a portable multimedia player. This is because digital images are nothing but a long sequence of zeroes and one's, making up the bits and bytes that only digital devices with digital processors recognise. Also, a digital camera is ultimately tasked with creating a digital image, which is made up of pixels—the smallest unit of any image whether still or motion picture. Images themselves very simply consist of light waves and their interaction with colours. Therefore a digital camera is tasked with capturing a scene (also called a subject) and converting this captured scene into pixel values. Of course capturing a scene is the big part of the job. Just like a traditional camera, a digital camera uses a set of lenses that focus light to create an image of a scene. While film cameras focus this light directly onto a piece of chemically reactive film, the lenses of a digital camera focus this light onto a semiconductor device (popularly called an image sensor) which is capable of storing light electronically.

Digital Photography touches each of our lives in some way today. Digital Cameras sell like hot cakes but do we know the ingredients behind the perfect image? Here's a look at how it's done...

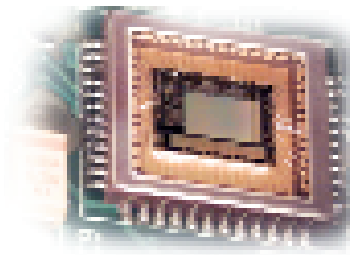
Cool Fact

The image sensor's ability to capture detail is measured in pixels or megapixels. A 7.1 megapixel camera displays 3072x2304 pixels or 7,077,888 pixels, when in fact it is supposed to display 7,100,000 pixels. This loss of a few pixels is attributed to the CCD circuitry. This loss isn't present in CMOS sensors.

Now that's all that there is to a digital camera. Did that sound easy? No? Good, because it's not. Let's look at the significant individual parts of a digital camera one at a time and list out their role (no pun) in creating a digital image from light.

The Image Sensor Photophobic And Loving It

We've already mentioned the role of the image sensor in capturing an image. Image sensors are basically silicon chips, (like your PC's CPU), around the size of your fingernail. The surface consists of millions of photosensitive diodes, each capable of capturing



An Image Sensor

a single pixel of the entire image (for ease of understanding you may assume each diode to be a pixel and the terms have been used interchangeably). These diodes convert the light that falls on them into electrons or an electrical charge. We all know when an image is clicked, the lens shutter opens briefly allowing light in. Now this light falls on each of these photosensitive diodes, (but with varying intensities according to the subject in the photograph), and is then converted into electrons. The more the light hitting a diode, the greater the charge it will record. For example, pixels capturing light from bright areas of a scene (bulbs, shiny objects, sunlight etc) will store a larger charge while those capturing light from dimmer areas (dark objects, shadows, etc) will have a lower charge. After the shutter closes to end the exposure, the charge from each pixel is measured and stored. From these numbers, result a digital image.

You will (surprisingly) not know that each diode can only capture brightness and not colour. Therefore, each pixel records an intensity of shade, a series of tones ranging from pure white to pure black. This is what we call *greyscale*. Wait now... where are all those gorgeous colours you see? That's the job of the next bad boy we're going to see—the Image Processor. But for now let's look at the two types of Image Sensors—CMOS (Complementary Metal Oxide Semiconductor) and CCD (Charge Coupled Device).

CCD sensors are generally smaller resulting in their use in mostly compact digital cameras. In a CCD sensor, every pixel's charge is transported across the chip to a single output node (the modern cameras may have more than one such node) where they are converted to analog voltages. This voltage is then buffered and sent off the chip as an analog signal. Then an ADC (Analog to Digital Converter) turns each pixel's analog value into a digital value by measuring the amount of charge present at that particular pixel's corresponding area.

CMOS sensors tend to be larger and till very recently were considered to be superior to CCDs which accounts for their prevalent use in DSLR (Digital Single Lens Reflex) a.k.a. professional cameras. On the CMOS sensor each pixel has its set of transistors that amplify, convert and move the charge. Often CMOS sensors also have additional circuitry for noise correction and digitalisation circuits, so that the sensor itself outputs digital bits rather than analog signals doing away



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